INCH-POUND

MIL-I-53011A(ME)
22 November 1988
SUPERSEDING
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31 August 1982

# MILITARY SPECIFICATION

# INTERFACE UNIT, ALARM

# J-4025/FSS-9(V)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers an alarm interface unit. It is a component of the joint-services interior intrusion detection system (J-SIIDS).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

| | |
|---|---|
| L-P-378 | - Plastic Sheet and Strip, Thin Gauge Polyolefin. |
| QQ-S-781 | - Strapping, Steel, and Seals. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-C-1752 | - Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible. |

MILITARY

| | |
|---|---|
| MIL-B-131 | - Barrier Material, Water Vaporproof, Flexible. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment. |
| MIL-STD-810 1/ | - Environmental Test Methods. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1190 | - Minimum Guidelines for Level C Preservation, Packing and Marking. |
| MIL-STD-45662 | - Calibration System Requirements. |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

| | |
|---|---|
| TA13226E2160 | - Interface Unit, Alarm J-4025/FSS-9(V) |

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE- JIE, Fort Belvoir, VA 22060-5606.)

1/ The "C" revision of MIL-STD-810 shall be used.



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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The alarm interface unit (hereinafter called "AIU") shall be in accordance with top assembly TA13226E2160 and as specified herein.

3.1.1 Function. The AIU is a printed circuit card which mounts inside the joint-services interior intrusion detection system (J-SIIDS) control unit. The AIU monitors the alarm/not alarm status of the control unit and conveys this status via double-pole double throw (Form C) relay output contacts to a commercial data transmitter and optional auxiliary device such as a siren. The AIU also provides an adjustable, direct current (dc) voltage for use by the commercial data transmitter.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the AIU. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provision of this specification.

3.3.1 Material deterioration prevention and control. The AIU shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the AIU may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components and shall make information available, upon request, to the contracting officer or designated representative.



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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the AIU may be newly fabricated from recovered materials to the maximum extent practicable, provided the AIU produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the AIU.

3.4 Environmental conditions. The AIU shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.1), performance degradation (see 6.4.2), or malfunction when tested as specified in 4.5.2.6.

3.4.1 Operational.

a. Temperature range of -20 to +150 °F .
b. Humidity up to 95 percent relative.

3.4.2 Nonoperational condition and storage.

a. Temperature range of -30 to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling.

3.5 Detailed technical requirements. The AIU shall perform as specified herein under any of the operating test conditions specified herein.

3.5.1 Terminal board functions. The terminal board (TB) functions shall be as follows:

| Terminal No(s) | Function |
|---|---|
| 1 | +20 Volts direct current (Vdc) Input. |
| 2 | Alarm/not alarm Input. |
| 3 | Adjustable 5 to 18 Vdc output. |
| 4 | Return for +20 Vdc input and adjustable output voltage. |
| 5-6 | Alarm/no-alarm output (relay contacts) |

| Terminal No(s) | Function |
|---|---|
| 5-7 | Alarm/no-alarm output (relay contacts) |
| 8-9 | Alarm/no-alarm output (relay contacts) |
| 8-10 | Alarm/no-alarm output (relay contacts) |



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3.5.2 Functional performance.

3.5.2.1 Input power. The AIU shall operate as specified below when provided with a regulated 20 ±1 Vdc at terminal 1, referenced to terminal 4. Current drain, when fully loaded and in not-alarm status, shall not exceed 200 ma.

3.5.2.2 Output power. The AIU shall provide at terminal 3, referenced to terminal 4, a regulated voltage adjustable from 5 to 18 Vdc, or a level 2 Vdc less than the supply voltage, at currents of up to 100ma.

3.5.2.3 Inputs. The AIU shall recognize the following input signals at the terminal board:

| Terminals | Voltage | State Indicated |
|---|---|---|
| 2-4 (2 positive) | less than 1 Vdc | Alarm |
| 2-4 (2 positive) | greater than 18 Vdc | Not alarm |

3.5.2.4 Outputs. Upon receipt of the input signals specified in 3.5.2.3, the AIU shall provide the following outputs at the terminal board:

| Input Signal | Terminals | Output Signals |
|---|---|---|
| Alarm | 5-6 | Relay closure |
| | 5-7 | Relay open |
| | 8-9 | Relay closure |
| | 8-10 | Relay open |
| Not alarm | 5-6 | Relay open |
| | 5-7 | Relay closure |
| | 8-9 | Relay open |
| | 8-10 | Relay closure |

3.5.2.5 Other indications. The AIU shall contain a light emitting diode (LED) which shall be illuminated when an alarm input is being received. The LED shall extinguish upon receipt of a not-alarm input signal.

3.6 Burn-in. Each AIU shall be subjected to a 24 hour burn-in to reduce the number of infant mortality failures (see 4.5.2.4).

3.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.8 Identification marking. The AIU shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.



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3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purposes intended.

3.9.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, requirement 5.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.7) for manufacturing of the AIU:

| Description | Identification | QTY |
|---|---|---|
| Stable base drawing of PWB | 13226E2161, sh 2 & 3 | 1 |

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance conformance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.3 Disassembly inspection. Failure of the first article to pass any examination or test shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in



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detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Government-loaned property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspection. The AIU shall be inspected as specified herein. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article AIU shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and for performing the disassembly inspection specified in 4.1.3.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the first article AIU shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection and for performing the disassembly inspection specified in 4.1.3.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Each AIU shall be examined for the major defects specified in 4.5.1. Presence of any defect shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each AIU shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column of 3 of table I. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The AIU shall be examined as specified herein for the following major defects:

Major

101. Assembly not correct.
102. Components missing or not as specified.
103. Dimensions not as specified.



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104. Material not as specified
105. Workmanship, including wiring and soldering, not as specified.
106 Identification or special marking missing, incomplete or illegible.
107. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
108. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
109. Contractor does not have documentation available for identification of material, material finish or treatment.
110. Used, rebuilt or remanufactured components, pieces or parts incorporated in the AIU.
111. Interchangeability not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Each AIU shall be subjected to a 24-hour burn-in as specified in 4.5.2.4. Unless otherwise specified herein, the AIU shall operate as specified herein without maintenance.

4.5.2.2 Test Schedule.

Table I. Test Schedule.

| First Article | Quality Conformance Individual | Sample | Test | Requirement Para | Test Para |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| X | X | - | Burn-in | 3.6 | 4.5.2.4 |
| X | X | - | Functional | 3.5.2.1-3.5.2.5 | 4.5.2.5 |
| X | - | X | Low Temp | 3.4.1(a), 3.4.2(a) | 4.5.2.6.1 |
| X | - | X | High Temp | 3.4.1(a), 3.4.2(a) | 4.5.2.6.2 |
| X | - | X | Humidity | 3.4.1(b), 3.4.2(b) | 4.5.2.6.3 |
| X | - | - | Shock | 3.4.2(c) | 4.5.2.6.4 |



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4.5.2.3 Test methods. Tests shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of the test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Burn-in. Each AIU shall be subjected to a 24-hour burn-in after successfully completing the functional test (see 4.5.2.5) and prior to performing further tests. During burn-in, the AIU shall be energized at 20.0 ±0.2 Vdc, adjusted to provide an output voltage of 5.0 ±0.3 Vdc into a load of 51 (±5%) ohms, and operated in the not-alarm state. Nine times every 24 hours, but not more than once per hour, the AIU shall be placed in the alarm state for not less than 10 seconds. Verification of the proper relay outputs, LED operation and the 5 Vdc output shall be made at these times. A complete functional test shall be made at completion of the 24 hour period. Non-compliance with the applicable portions of 3.5 shall constitute a failure. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order (see 6.2), burn-in test reports shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional 24-hour burn-in after failure analysis and repair and successful completion of the functional test.

4.5.2.5 Functional. The AIU shall be tested in accordance with Appendix test method 1.

4.5.2.6 Environmental. The AIU shall be subjected to the following series of tests specified in MIL-STD-810. During each test, conduct the functional test specified in 4.5.2.5. At the completion of the environmental tests, conduct the functional tests specified in 4.5.2.5.

4.5.2.6.1 Low temperature. Conduct low temperature test method 502.1, procedure I, storage temperature of -30 °F for not less than 48 hours, equipment operating temperature of -20 °F.

4.5.2.6.2 High temperature. Conduct high temperature test method 501.1, procedure II, except that the storage temperature during step 4 shall be +165 °F instead of the operating temperature shall be +150 °F.

4.5.2.6.3 Humidity. Conduct test method 507.1, procedure I with the following exceptions:

a. Step 5 shall be for 5 cycles (not less than 120 hours). Perform the functional test at the completion of each cycle.
b. During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
c. During step 4, the relative humidity shall be maintained at 92 percent or greater.
d. Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle.

4.5.2.6.4 Shock. Bench handling: Conduct method 516.2, procedure V.



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4.5.2.6.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

| | Defect | Req. Paragraph |
|---|---|---|
| 112. | AIU's not wrapped and ends sealed as specified for level A or B. | 5.1.1, 5.1.2 |
| 113. | AIU's not placed in bag, excessive air exhausted, and bag heat sealed as specified for level A or B. | 5.1.1, 5.1.2 |
| 114. | AIU's not preserved and packed in accordance with the referenced document for level C. | 5.1.3, 5.2.3 |
| 115. | AIU's packed together exceed the specified limitations as specified for level A or B. | 5.2.1, 5.2.2 |
| 116. | Strapping not as specified for level A | 5.2.1 |
| 117. | Marking missing, illegible, incorrect, or incomplete. | 5.3 |

5. Packaging.

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each AIU shall be wrapped in polyurethane cushioning material conforming to PPP-C-1752, type II, class 1, 1/4 inch thick. The cushioning shall be of sufficient size to allow the ends of the material to be stapled or secured with tape to protect the ends of the AIU from protruding. The wrapped AIU shall then be placed in a close-fitting heat-sealed bag, fabricated from material conforming to MIL-B-131, class 2. Included air volume shall be kept at a minimum. The AIU shall then be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the box specification.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for level A except the bag shall be fabricated from material conforming to L-P-378 with a minimum nominal 4 mils thickness.



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5.1.3 Level C. Each AIU shall be preserved in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, level B, or C as specified (see 6.2).

5.2.1 Level A. AIU's, preserved as specified in 5.1, shall be packed together in a close-fitting box confirming to PPP-B-601, overseas type, style optional, in quantities not to exceed 200 pounds gross weight. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Strapping shall be finish "B".

5.2.2 Level B. AIU's preserved as specified in 5.1 shall be packed same as for level A in 5.2.1, except that boxes shall be domestic type and the strapping may be finish A.

5.2.3 Level C. AIU's preserved as specified in 5.1 shall be packed in shipping containers in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special or identification markings required by the contract or purchase order and 5.3.1, all containers shall be marked in accordance with MIL-STD-129.

5.3.1 Additional marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the project code "JID".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AIU is intended for mounting inside the J-SIIDS control unit. It is used to reformat the alarm output signal to interface with commercial data transmitters.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2..1.1 and 2.2).
c. Revision level of Technical Data Package List to which the AIU is to be built and any changes thereto.
d. Serial numbers and method of sequencing to be used.
e. When a first article is required for inspection and approval, the time frame required for submission of the first articles, and the number of units required (see 3.2).
f. When Government-loaned property is not required (see 3.10).
g. Level of preservation and packing required (see 5.1 and 5.2).
h. Any special marking (see 5.3).
i. Schedule, content, and format of burn-in failure reports (see 4.5.2.1).



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6.3 First article. When a first article inspection is required, the items should be a preproduction AIU. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the AIU, evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.4.2 Performance degradation. A change in electrical or mechanical characteristics to such a degree that when subjected to test method 1 in Appendix A, the AIU fails to meet one or more requirements of 3.5.

6.5 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and test performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.7 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.10 when applicable (see 6.2).

6.8 Subject term (key word) listings.

Alarm system
Intrusion detection
Physical security



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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 6350-A156



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# APPENDIX

# TEST METHOD

# FOR

# INTERFACE UNIT, ALARM

# J-4025/FSS-9(V)

## 10. SCOPE

10.1 Scope. The test methods contained in this Appendix are to determine whether AIU's procured under this specification conform to the requirements. Figure A-1 contains data for fabrication of a test set which is required to facilitate testing of the AIU.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.



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APPENDIX

TEST METHOD 1

FUNCTIONAL TEST

1. TEST APPARATUS:

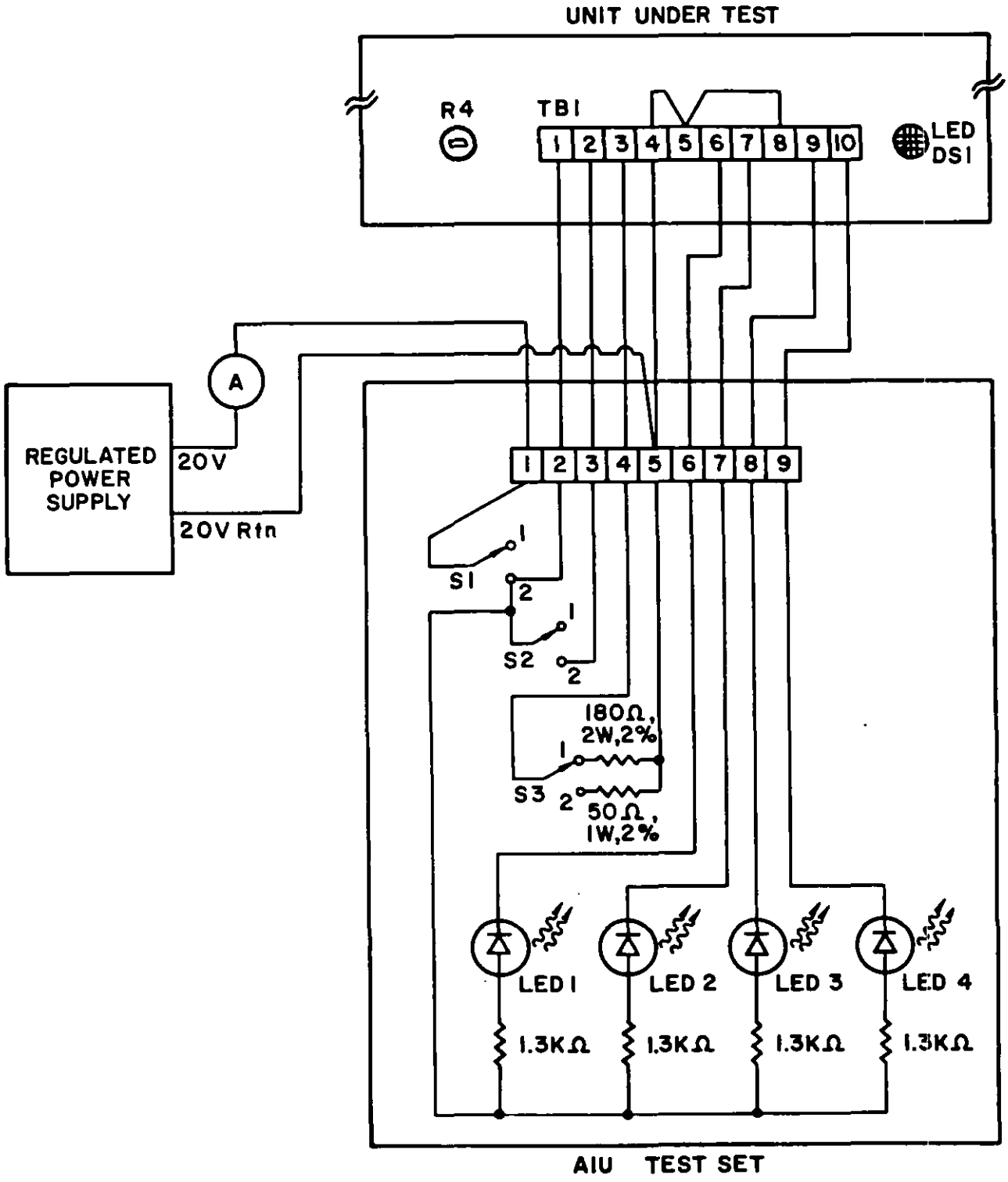
a. Test set - see figure A-1.
b. Regulated power supply - HP6202 or equivalent.
c. 2 ea: Digital Multimeters - Fluke 8030A or equivalent.

2. TEST PROCEDURE:

a. Connect AIU under test to test apparatus as shown in figure A-1, with all switches set to position 1 as shown.
b. Set power supply to +20 ±.2 Vdc.
c. Place switch S1 in position 2. LED's 1, 3, and DS1 shall be lit, and LED's 2 and 4 shall be unlit.
d. Place S2 in position 2. LED's 2 and 4 shall be lit and all other LED's unlit.
e. Adjust R4 to provide an output voltage of 18 ±.5 Vdc across TB1-3 and TB1-4, except, when performing test during temperature and humidity testing, no adjustment of R4 is permitted. In those tests, voltage reading shall not differ by more than 0.2 Vdc from that of step 2j in the last previous functional test.
f. Verify that the current through the ammeter is less than 200 ma.
g. Adjust R4 to provide output voltage of 5 ±0.3 volts at TB1-3.
h. Place S3 in position 2. Voltage shall change by no more than 0.2 Vdc.
i. Verify that the current through the ammeter is less than 200 ma.
j. If temperature or humidity tests are being or may be performed later, place S3 in position 1 and adjust R4 for voltage at TB1-3 of 18 ±0.2 Vdc. Record this voltage for later comparison.



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1. ALL RESISTORS ARE 1/2 WATT, 10% TOLERANCE UNLESS OTHERWISE NOTED.
2. LED'S 1 THRU 4 ARE MONSANTO MV 5053 OR EQUIVALENT.

FIGURE A-1. Test fixture schematic and interconnecting wiring diagram.

X-4091A



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-I-53011A(ME) | Interface Unit, Alarm J-4025/ESS-9(V) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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